

Day 3 - Mentor Review

1. Introduction Problem Statement : Underwater Optical Wireless Communication (UWOC)

In UWOC, information is transmitted as binary bits (0 and 1) using optical signals. As light travels through water, turbidity (NTU) causes absorption and scattering, which changes the received optical signal amplitude. These variations make fixed-threshold detection unreliable and can lead to bit errors. To overcome this challenge, supervised machine learning models are used to classify transmitted bits based on received signal values and turbidity information, enabling more robust communication under varying underwater conditions.

2. Solution for this : ML

- Domain: Underwater Optical Wireless Communication (UWOC)
- Problem Type: Binary Classification
- Learning Type: Supervised Machine Learning
- Target Classes: Bit 0 and Bit 1

3. Objectives :

- Understand the UWOC problem and binary bit classification (0/1).
- Analyze received signal and NTU/turbidity data.
- Build a preprocessing and exploratory data analysis (EDA) pipeline.
- Implement and compare Fixed Threshold, Logistic Regression, SVM, and Random Forest.
- Evaluate using Accuracy, Precision, Recall, F1-Score, BER, and Inference Time.
- Test robustness across different NTU levels and identify limitations.



4. Tasks for 4 weeks :

- Problem Understanding & Literature Review
- Dataset Understanding & Quality Assessment
- Data Preprocessing & Exploratory Analysis
- Feature / Representation Preparation
- Baseline & Main Model Development
- Tuning & Project-Specific Analysis
- Validation, Error & Failure Analysis
- Documentation & Presentation